

Hotel Bookings Analysis

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1. Key Observations

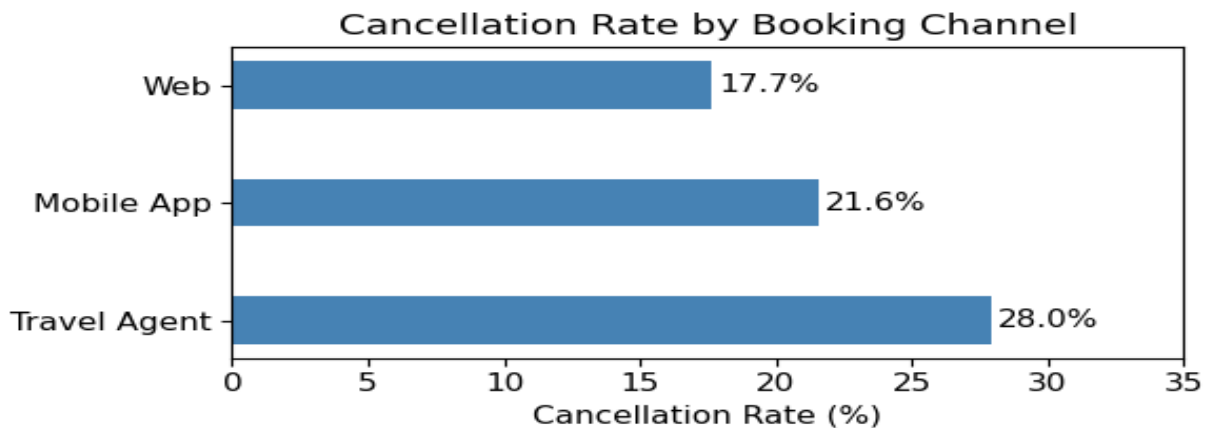
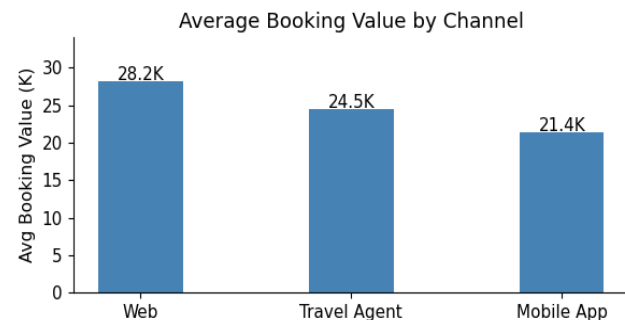
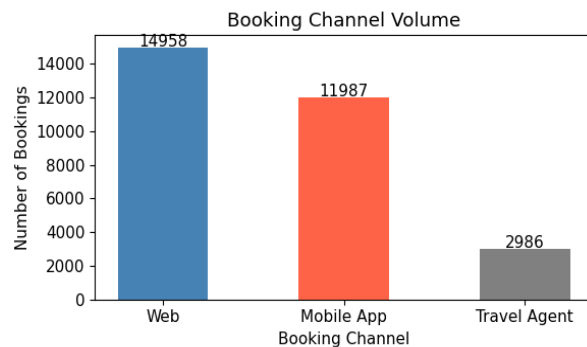
Observation 1: Web drives the best quality demand.

Web contributes about **50% of bookings**, Mobile App about **40%**, and Travel Agent about **10%**. But performance is not equal:

- **Web cancellation rate: 17.7%**
- **Mobile App cancellation rate: 21.6%**
- **Travel Agent cancellation rate: 28.0%**

Web also has the **highest average booking value (~28.2K)**, well above:

- Mobile App: **~21.4K**
- Travel Agent: **~24.5K**
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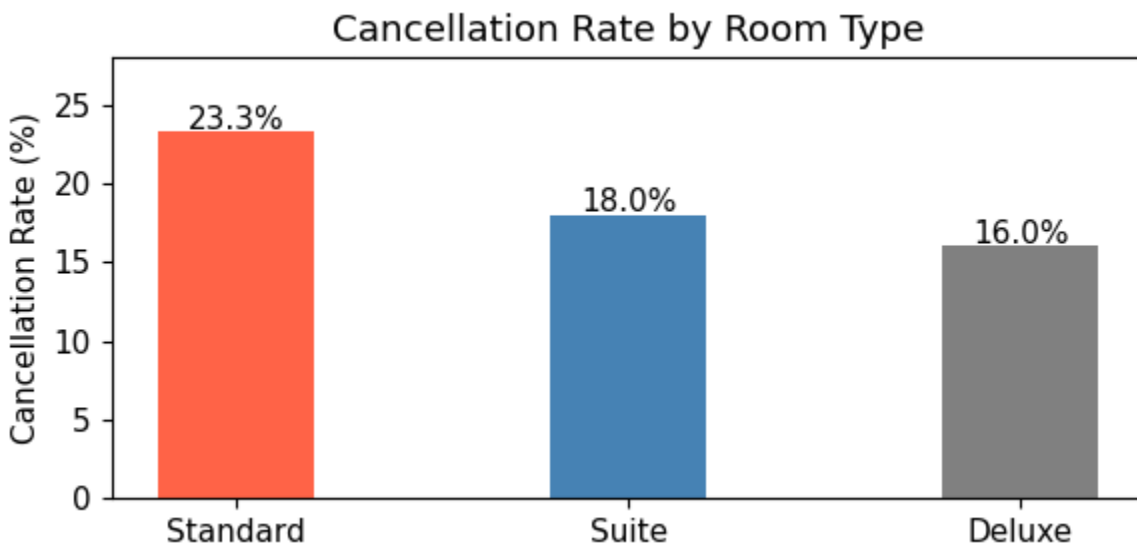
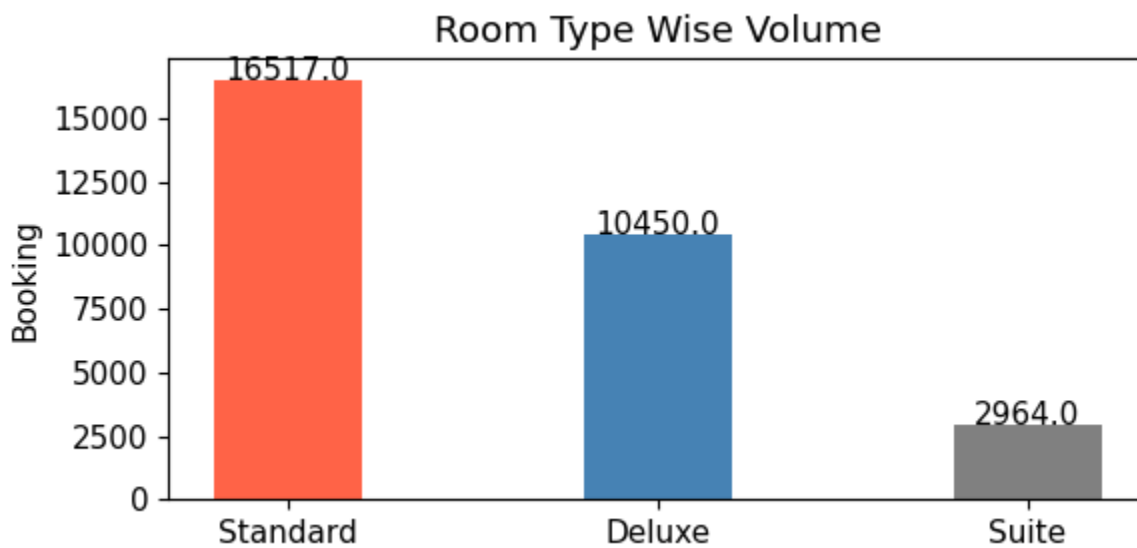
Observation 2: Standard rooms bring volume, but also the biggest cancellation risk.

Room mix is heavily concentrated in:

- **Standard: 55%** of bookings
- **Deluxe: 35%**
- **Suite: 10%**

Cancellation by room type:

- **Standard: 23.3%**
- **Suite: 18.0%**
- **Deluxe: 16.0%**

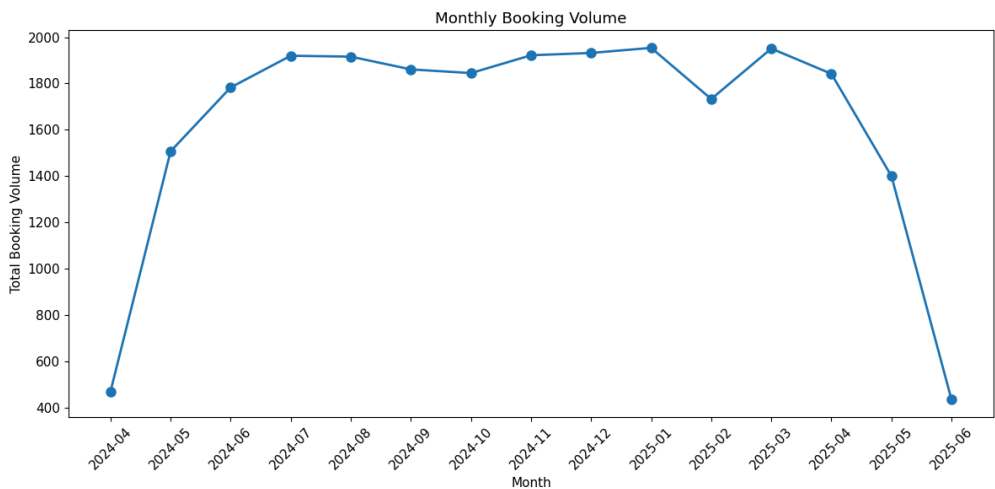
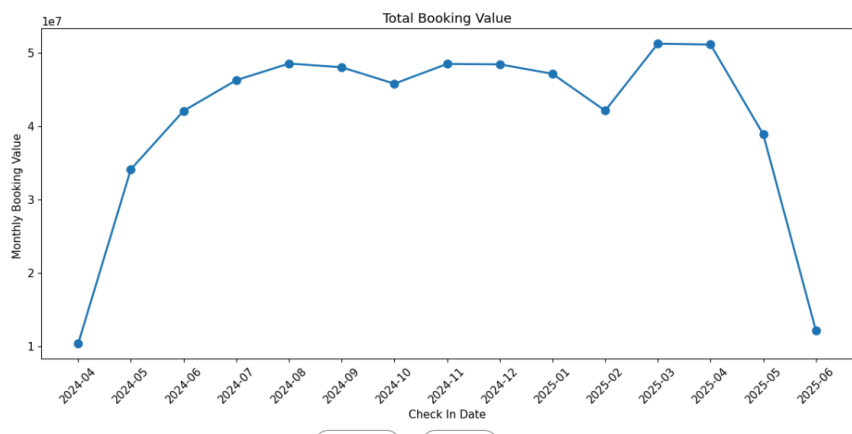
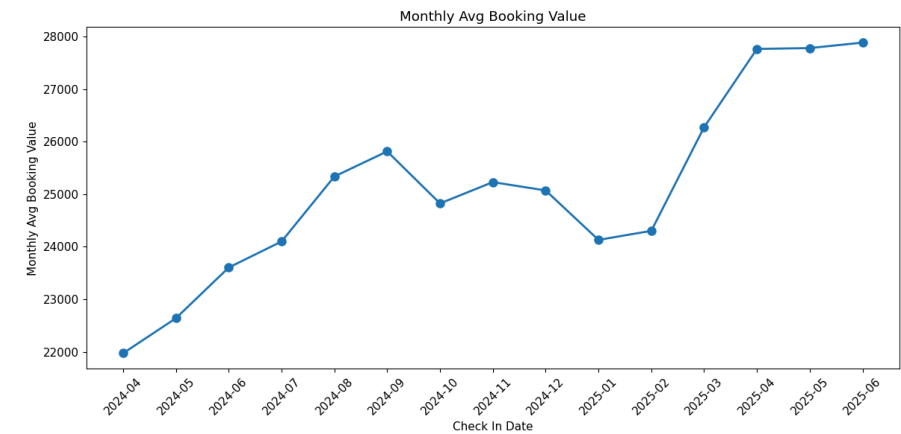


Observation 3: Booking value is rising, but stay length is not.

Average booking value by check-in month rises from about **21.9K in Apr 2024** to **27.8K by Apr 2025** and **28.2K by Jun 2025**.

At the same time, **average stay length stays almost flat at ~4 days** , **booking value decreases** and **volume decreases**.

What this means: value growth is based on increase in pricing due to pricing as average booking price stays strong.



1.2 How booking patterns vary across channels, room types, and star ratings

By channel

- **Web:** highest booking value, lowest cancellation
- **Mobile App:** okay booking value, weaker conversion
- **Travel Agent:** smallest volume, worst cancellation profile

By room type

- **Standard:** mass-market, high volume, highest cancellations
- **Deluxe:** best balance of scale and stability
- **Suite:** niche segment, lower volume, moderate cancellations

By star rating

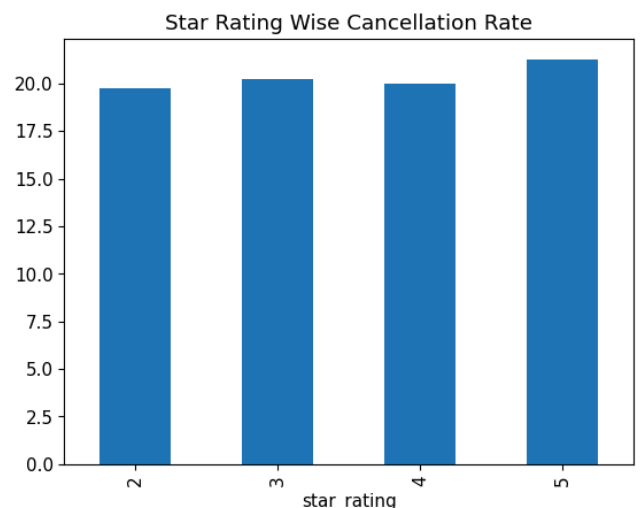
Booking concentration is strongest in:

- **4-star:** ~40%
- **3-star:** ~35%
- **5-star:** ~15%
- **2-star:** ~10%

Cancellation rates across star ratings are surprisingly similar:

- **5-star:** 21.3%
- **3-star:** 20.2%
- **4-star:** 20.0%
- **2-star:** 19.8%

I feel Star Rating is not the main driver as cancellation rate is almost the same as I feel room type and channel contribute majorly.



2. Root Cause Analysis

What might be causing the cancellation patterns?

- Agents sometimes make speculative bookings to lock in rates, intending to cancel if plans fall through
- Standard rooms are the cheapest option, which invites more casual or impulse bookings. There is also a high supply of standard-tier rooms, so guests feel they can always find another option if they change their mind as they get a better deal. [Price Sensitive]
- The flat structure across all months means the business is not capturing the extra willingness to pay during peak periods. Revenue growth is being driven purely by volume, not price optimization.
- The spikes in **Jul–Aug** and **Dec–Jan** point to holiday / peak travel volatility which increases cancellation due to competitive price war , increase in hotel prices

Why some booking channels or property types perform better than others

Web outperform because

- No middleman involved
- lower cancellation
- higher booking value
- High Intent

Mobile underperform because

- booking values are lower
- cancellations are higher than web
- Brings volume but not quality

Deluxe seems to sit in the sweet spot:

- good volume
- lowest cancellation
- likely less price sensitive than Standard
- broader appeal than Suite

Are there seasonal or temporal trends influencing booking values or stay lengths

Yes, clearly.

Booking value trend

- steadily rises across the year
- strongest levels show up in late 2024 and 2025

Stay length trend

- almost unchanged at ~4 days

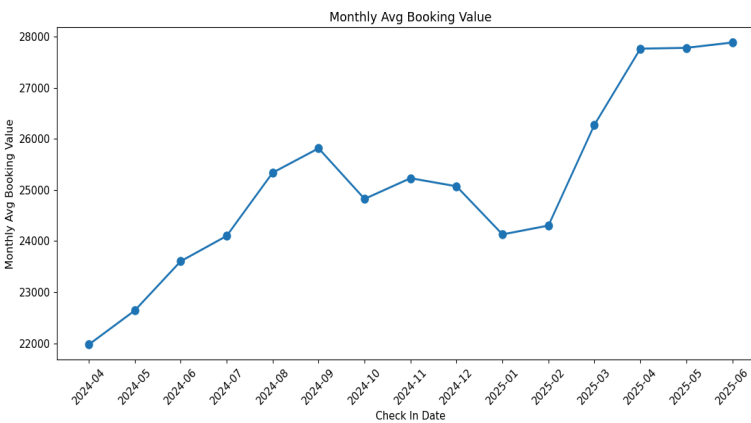
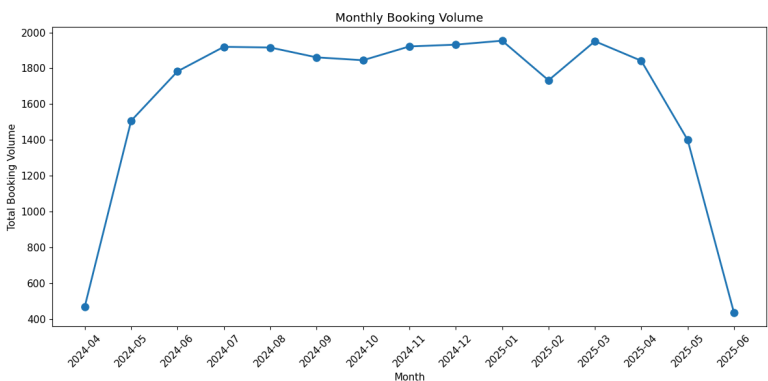
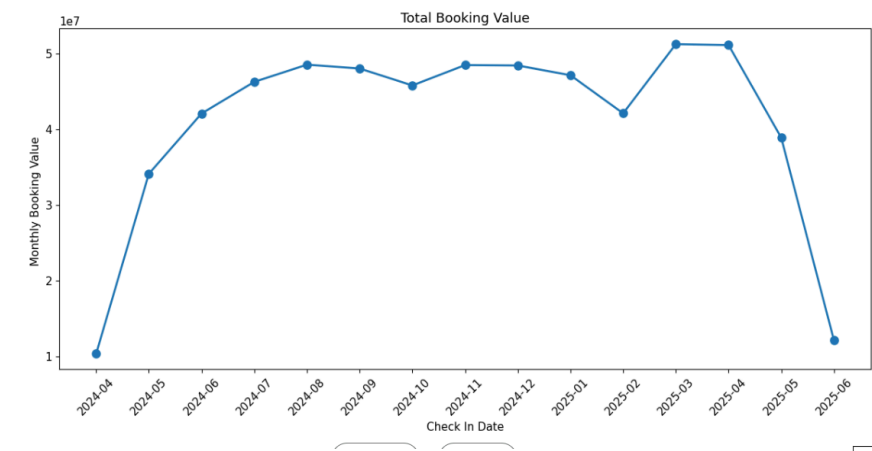
Interpretation

- the business is extracting more value per booking
- but this is not coming from longer stays
- it is more likely being driven by pricing or seasonality

Monthly Average length of stay per month
year_month

2024-04	4.127660
2024-05	4.037823
2024-06	3.988783
2024-07	4.030208
2024-08	4.039144
2024-09	4.013434
2024-10	3.982114
2024-11	3.965661
2024-12	4.060559
2025-01	3.928864
2025-02	4.010387
2025-03	4.015377
2025-04	4.002714
2025-05	3.996429
2025-06	3.981567

So the business is winning on **extracting more value per booking**, but not yet on **stay extension**.



3. Business Recommendations

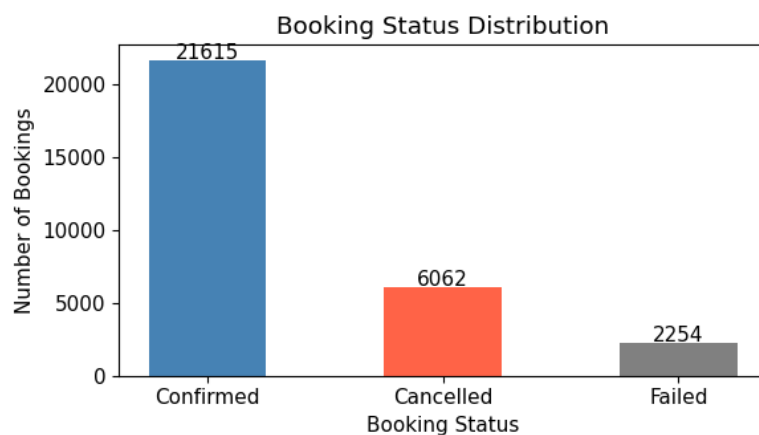
Strategies to reduce cancellations

Tighten cancellation policy

- Apply stricter rules for **Travel Agent** and **Standard room** bookings
- Introduce partial prepayment for high-risk segments
- Can do like Flipkart give them 5-10% discount extra and put a non refundable policy which creates and filters intent of customers(For Flexibility free date adjustment).
- Provide them free breakfast .

Ways to improve profitability and repeat bookings

- Push more traffic to the Web , low cancellation ,high booking price
- Give upgrades to people from standard to Deluxe while booking which reduces the risk of cancellation
- According to data in a year Avg repeat booking is 60 so we can have Points loyalty programs where on travelling with us they will get points and cashbacks that can be redeemed on next trip.
- shifting acquisition budget toward Web direct bookings
- Work on interface and APIs as 2254 payments are failed
- Increase prices in peak months, but pair them with tighter cancellation terms



```
total_order_per_user=hotel['customer_id'].value_counts()
repeat_cus=(total_order_per_user>1).count()
avg_repeat_bookings=total_order_per_user[total_order_per_user>1].mean()
total_user=len(total_order_per_user)
print("Average Repeat Bookings are ",avg_repeat_bookings)
```

Average Repeat Bookings are 59.98196392785571

